

Erdős Problem #680

EP-680

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Is it true that, for all sufficiently large n , there exists some k such that $p(n+k) > k^2 + 1$, where $p(m)$ denotes the least prime factor of m ?

Can one prove this is false if we replace $k^2 + 1$ by $e^{(1+\epsilon)\sqrt{k}} + C_\epsilon$, for all $\epsilon > 0$, where $C_\epsilon > 0$ is some constant?